The database diagrams, explained simply

A plain-English guide to erd-platform.md and erd-ai-mcp.md.

First, how to read the lines

A **table** is a list of one kind of thing, like a list of users or a list of rooms.

A **relation** is a link between two tables. Almost every link in this diagram is the same kind:

> **"One A has many B."** For example, one business has many rooms, and each room belongs to one business.

Some links are **optional**. For example, a booking *may* have a user, because guests can book without an account.

A few links are **one-to-one**. For example, one booking has one review at most.

Part 1: The Platform ERD

👤 People and logging in

1. User

A person with an account. It could be a guest, a business owner, a staff member, or a platform admin. `platformRole` says if they are a normal user or an admin of the whole site.

- One user → **one host profile** (at most)
- One user → **many sessions**
- One user → **many accounts**
- One user → **many memberships** (they can work at many businesses)
- One user → **many bookings** and **many session bookings** (optional, since guests can book without logging in)
- One user → **many favorites**, **many reviews**, **many activity logs**
- One user → **many plan requests** (the person who asked)
- One user → **many push subscriptions**
- One user → **many AI chats**, **API keys** and OAuth things (explained in Part 2)

2. HostProfile

The "about the host" card shown to guests: name, bio, city, languages, WhatsApp, and how to contact them.

- It belongs to **one user**. A user has at most one.

3. Session

"You are logged in right now on this device." When it ends, you are logged out.

- It belongs to **one user**.
- It also has `mcpBusinessId`: when an AI app is logged in, this says which business it is working on.

4. Account

How a user logs in: with a password, or with Google, and so on.

- It belongs to **one user**. One user can have many ways to log in.

5. Verification

Short-lived codes, for example "verify your email" or "reset your password." It has no links.

🏨 Businesses and teams

6. Business

The main thing on the site: a guesthouse, a hotel, a chalet, a tour company. It holds the name, address, map location, currency, check-in time and house rules. It also holds the business's **choices**:

- `bookingMode`: how guests book (for example instantly, or after the owner says yes)
- `paymentCollection` and `onlineAmount`: whether guests pay online or at the place, and how much
- `refundPolicy`: the rules for money back
- `plan`: which subscription the business pays for
- `status`: for example active or suspended

The business is the "parent" of almost everything:

- One business → **many** roles, memberships, rooms, photos, bookings, occupancies, calendar feeds, payments, refunds, payouts, plan payments, plan requests, reviews, favorites, offerings, sessions, session bookings, stored images, activity logs, API keys and MCP logs.

7. Role

A job title inside a business, like "Owner" or "Receptionist," with a list of what it is allowed to do (`permissions`).

- It belongs to **one business**.
- One role → **many memberships**.

8. Membership

"This person works at this business with this role." It **connects** a user, a business and a role.

- It belongs to **one user**, **one business** and **one role**.
- One membership → **many API keys**.

🛏️ Rooms, photos, prices

9. Room

A room or unit that can be booked: how many guests, bedrooms, beds, bathrooms, the normal price per night, and amenities like Wi-Fi or a jacuzzi.

- It belongs to **one business**.
- One room → **many** photos (optional), price changes, calendar feeds, bookings, occupancies and reviews.

10. Photo

A picture. It always belongs to **one business**. It can also belong to **one room** *or* **one offering**.

11. RateOverride

"On this date, this room costs a different price." For example, a higher price on New Year's Eve.

- It belongs to **one room**.

12. StoredImage

The real image file, kept inside the database.

- It belongs to **one business**.

🏷️ Categories

13. Category

A type of place or activity, like "Chalet," "Guesthouse" or "Hiking," with an English name, an Arabic name and an icon.

14. BusinessCategory

A **connector table**. One business can be in many categories, and one category can hold many businesses. Each row just says "business X is in category Y."

- It belongs to **one business** and **one category**.

🗓️ Bookings and calendars

15. Booking

A guest booking a **room** for some nights. It holds the guest's name and phone, check-in and check-out dates, the price, the deposit and the status (waiting, confirmed, cancelled…).

- It belongs to **one business** and **one room**.
- It *may* belong to **one user**.
- One booking → **one occupancy** (at most)
- One booking → **many payments**
- One booking → **one review** (at most)

16. Occupancy

"These nights are taken." This is the room's calendar. A night can be taken for three reasons:

1. a booking on our site
2. the owner blocked it by hand
3. it came from Airbnb or Booking.com through a calendar feed

A database rule makes sure **two stays can never take the same room on the same night**. No double booking.

- It belongs to **one business** and **one room**.
- It *may* belong to **one booking** (one-to-one).
- It *may* belong to **one calendar feed**.

17. CalendarFeed

A link to an **outside calendar** (Airbnb, Booking.com) for one room. The site reads it, so nights booked there also show as taken here.

- It belongs to **one business** and **one room**.
- One feed → **many occupancies**.

🧗 Experiences (tours and activities)

18. Offering

An activity, not a room. For example a hike, a cooking class or a boat tour. It holds the price, how long it takes, the meeting point and the level (easy or hard).

- It belongs to **one business**.
- It *may* belong to **one category**.
- One offering → **many** sessions, session bookings, photos and reviews.

19. OfferingSession

One date and time for an offering, with a number of seats. For example "Saturday 9am, 12 seats."

- It belongs to **one business** and **one offering**.
- One session → **many session bookings**.

20. SessionBooking

A guest booking **seats** in one session. It is like Booking, but for activities. A database rule makes sure a session **never sells more seats than it has**.

- It belongs to **one business**, **one offering** and **one session**.
- It *may* belong to **one user**.
- One session booking → **many payments**.
- One session booking → **one review** (at most).

💰 Money

21. Payment

Money a **guest** paid, online or in cash.

- It belongs to **one business**.
- It belongs to **either** one booking **or** one session booking.
- One payment → **many refunds**.

22. Refund

Money given **back** to a guest.

- It belongs to **one business** and **one payment**.

23. Payout

The business asks the platform: "Send me my money," by bank or by Whish.

- It belongs to **one business**.

24. PlanPayment

The business paying **online** for its subscription plan.

- It belongs to **one business**.

25. PlanRequest

The business paying for its plan **by hand**, for example by bank transfer with a receipt photo. An admin then says yes or no.

- It belongs to **one business**.
- It *may* belong to **one user** (the person who asked).

26. PlatformPaymentOption

The ways **the platform** accepts manual plan payments, like bank details. It has no links.

27. WhishSandboxPayment

Fake Whish payments, for testing only. It has no links.

28. WebhookEvent

Messages that payment companies send to us ("this payment succeeded"). We save each one so we never handle the same message twice. It has no links.

⭐ Guests' extras

29. Favorite

"This user saved this business with a heart."

- It belongs to **one user** and **one business**.

30. Review

A star rating and a comment. The owner can reply, and an admin can hide it.

- It always belongs to **one business**.
- It *may* belong to a room, a booking, an offering, a session booking and an author (user).
- One booking or one session booking has **one review at most**.

🔧 System tables

31. ActivityLog

A diary of "who did what," for example "Sara confirmed booking #123."

- It *may* belong to **one user** and *may* belong to **one business**.

32. PushSubscription

Lets us send **notifications** to a phone or browser.

- It *may* belong to **one user**. A guest without an account can still get updates through `bookingToken`.

33. SiteSetting

Settings for the whole site, like the color theme. It has no links.

34. RateLimitWindow

Counts how many requests someone makes, to stop spam. It has no links.

Part 2: The AI and MCP ERD

MCP is the way outside AI apps (like Claude or ChatGPT) connect to a business's data. **OAuth** is the safe "Log in and allow this app" process.

🤖 The site's own AI assistant

1. AiChat

One conversation with the AI assistant, with a title.

- It belongs to **one user**.
- It *may* belong to **one business** (when an owner chats about their business).
- One chat → **many messages**.

2. AiChatMessage

One message in a chat, from the user or from the AI.

- It belongs to **one chat**.

3. AiUsage

How much each AI answer **cost**: tokens and money. It has no real links. It just notes the business id as plain text.

4. JobProgress

Live status of slow background work, like "searching rooms…", so the screen can show progress while the answer is being made. It has no links.

🔑 Outside AI apps: simple keys

5. ApiKey

A **secret key** a staff member makes so an AI app can reach the business's data. It can be read-only or allowed to change things (`canWrite`). It can expire or be turned off. We only save a **hash** of the key, never the key itself.

- It belongs to **one business**, **one membership** and **one user** (who made it).
- Because it is tied to the membership, if the person leaves the business, the key stops being useful.

6. McpAuditLog

A diary of **every action an AI app did**: which tool, whether it worked, and how fast. It is for safety and checking.

- It belongs to **one business**.

🔐 Outside AI apps: "Log in and allow" (OAuth)

Think of it like a hotel: the **app** is a visitor, the **consent** is the owner saying "let them in," the **access token** is a day pass, and the **refresh token** lets them get a new day pass without asking again.

7. OauthClient

An **app** that is registered to connect, for example Claude. It holds its name, logo, and where to send people back after login.

- It *may* belong to **one user**.
- One app → **many** client-resources, refresh tokens, access tokens and consents.

8. OauthResource

The thing being protected: our MCP server. It also says how long passes last.

- One resource → **many client-resources**.

9. OauthClientResource

A **connector table**: "this app may use this resource."

- It belongs to **one app** and **one resource**.

10. OauthConsent

"This user said **yes**, this app may access these things (`scopes`)."

- It belongs to **one app**. It *may* belong to **one user**.

11. OauthAccessToken

The **short pass** the app shows on every request.

- It belongs to **one app**.
- It *may* belong to **one user**, **one session** and **one refresh token** (the one that made it).

12. OauthRefreshToken

The **long pass** used to get new short passes.

- It belongs to **one app** and **one user**. It *may* belong to **one session**.
- One refresh token → **many access tokens**.

13. Jwks

The site's **signing keys**. They are like a stamp that proves a pass is real. It has no links.

14. OauthClientAssertion

Remembers one-time proofs an app already used, so nobody can use them again. It has no links.

Not in the diagram: the job queues for AI, emails and schedules. A tool called pg-boss keeps them in its own part of the database.